Gene co-expression modules

WGCNA

What is “gene co-expression module”?

- A co-expression module is a set of genes with similar expression patterns across samples, typically identified using correlation-based clustering methods
- Genes rarely act alone. Many genes are turned on and off together because they participate in the same biological process (e.g., cell cycle, immune activation, metabolism)
- If gene A is high whenever gene B and C are high, and low when they are low → they likely belong to the same module

How are co-expression modules identified?

1. Compute gene-gene similarity
    a. Usually Pearson correlation:
    b. High correlation → similar expression patterns.
2. Build a gene network
    a. Each gene = node
    b. Edge weight = correlation between genes
    c. Instead of hard thresholds, modern methods use weighted networks
3. Detect modules (clusters)
    a. Correlations are transformed into connection strengths
    b. Genes are grouped into modules using hierarchical clustering
4. Summarize each module
    a. Each module is summarized by its module vector
    b. This is a single vector represents module’s overall activity across samples

Example 1

Gene \ Sample	Sample 1	Sample 2	Sample 3	Sample 4
Gene1	5.1	6.2	5.9	6.5
Gene2	4.9	6	6.1	6.3
Gene3	0.2	0.1	3.3	2.6

Genes 1 and 2 clearly move together → likely same module.

Example 2:

Analysis identified:

Blue module → 150 genes

Brown module → 80 genes

Blue module increases in diseased samples → suggests disease relevance.

Why correlate modules instead of individual genes in multiomics?

- Instead of testing 30,000 genes × millions of SNPs, we test:
  - Modules (10–50) × SNPs
- Advantages:
  - Reduced multiple testing
  - More biologically interpretable
  - Captures coordinated gene regulation

How do we correlate modules with variants?

1. Use module vector
   a. Each sample has:
   b. One value per module (module eigengene)
   c. One genotype value per SNP
2. Association testing
   a. For each SNP and each module:
3. This is typically:
   a. Linear regression
   b. Pearson correlation
   c. Linear mixed models (for population structure)
4. We obtain module QTL (modQTL) or co-expression QTL

Individual	SNP rs1234	Variant	Module vector
Person 1	AA	0	0.02
Person 2	AG	1	8.1
Person 3	GG	2	2.3
Person 4	AA	0	0.03
Person 5	GG	2	2.2

1. Means:

$$\bar{x} = 1, \quad \bar{y} = 2.53$$

2. Sample covariance:

$$\mathrm{cov}(x, y) = 1.1125$$

3. Standard deviations:

$$\sigma_x = 1, \quad \sigma_y \approx 3.306$$

4. Pearson correlation coefficient:

$$r = \frac{\mathrm{cov}(x, y)}{\sigma_x \sigma_y} = \frac{1.1125}{1 \cdot 3.306} \approx 0.34$$

Introduction to WGCNA (Weighted Gene Co-expression Network Analysis)

Definition: A systems biology method for describing correlation patterns among genes across microarray or RNA-seq samples - identify modules of highly co-expressed genes.

Key Features:

- Builds gene co-expression networks based on pairwise correlations
- Uses a "soft" thresholding power to emphasize strong correlations and penalize weak ones
- Groups genes into modules (clusters) based on similar expression patterns
- Relates modules to external sample traits (e.g., disease status, clinical data)
- Identifies hub genes - highly connected genes within modules

What can WGCNA tell us?

"Which sets of genes act together,

and which genetic variants regulate those gene programs?"

WGCNA - simple example

WGCNA on a Simple Example: From Data to Insight

Small dataset, concrete numbers, clear results

1. The data
6 samples, 8 genes
+ Genotypes & trait

Gene expression (log2)

Sample	Gene1	Gene2	Gene4	Gene5	Gene6	Gene7	Gene8
S1	5.1	5.3	7.0	7.6	7.7	3.1	3.0
S2	5.0	5.2	4.8	8.0	7.9	3.3	3.1
S3	4.9	5.0	5.1	7.7	7.4	3.2	3.2
S4	1.6	1.8	1.7	3.0	3.1	2.9	8.0
S5	1.7	1.9	1.6	2.9	3.0	8.1	7.9
S6	1.5	1.6	1.4	3.1	3.2	3.8	7.7

Genotype (SNP1: 0/1)

S1	S2	S3	S4	S5	S6
0	0	0	1	1	1

Trait (Disease status)

S1	S2	S3	S4	S5	S6
0	0	0	1	1	1

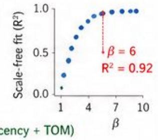
(0 = healthy, 1 = diseased)

2. Build the network (WGCNA steps)

Compute correlations between genes
Example: cor(Gene1, Gene2) = 0.99
cor(Gene1, Gene7) = -0.98

↓

Choose soft-threshold power (β)
We test β = 1...10 and look for scale-free topology
Best β = 6 (R^2 = 0.92)



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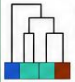
Construct network (adjacency + TOM)
Stronger connections within groups, weaker between groups

↓

Cluster genes & detect modules



Result: 2 modules (+ grey for none)
– Module Blue: Gene1, Gene2, Gene3
– Module Turquoise: Gene4, Gene5, Gene6
– Module Brown: Gene7, Gene8

3. Module eigengenes (summary of each module)

First principal component of each module

Module eigengene (ME)

Sample	MEblue	MEturquoise	MEbrown
S1	1.07	1.05	-1.11
S2	1.02	1.12	-1.05
S3	1.08	1.00	-1.08
S4	-1.06	-1.02	1.10
S5	-1.02	-1.05	1.07
S6	-1.09	-1.10	1.07

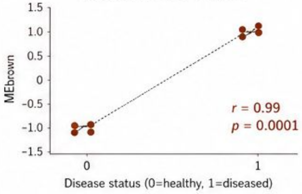
Modules capture the main patterns:
Blue & Turquoise high in healthy (S1–S3), low in diseased (S4–S6).
Brown shows the opposite.

4. Relate modules to trait

Correlate module eigengenes with disease status

Module	Correlation (r)	p-value
Blue	-0.98	0.0003
Turquoise	-0.98	0.0003
Brown	0.99	0.0001

MEbrown vs. Disease status



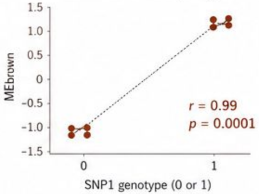
Brown module is strongly associated with disease.

5. Relate modules to genotype (eQTL-style insight)

Correlate module eigengenes with SNP1 (0/1)

Module	Correlation (r)	p-value
Blue	-0.98	0.0003
Turquoise	-0.98	0.0003
Brown	0.99	0.0001

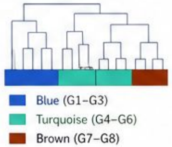
MEbrown vs. SNP1 genotype



Interpretation:
SNP1 genotype is highly associated with the Brown module.
Genes in this module (Gene7, Gene8) are likely under genetic regulation by SNP1.

6. What did we learn?

Network structure
WGCNA grouped genes into 3 modules based on co-expression patterns.



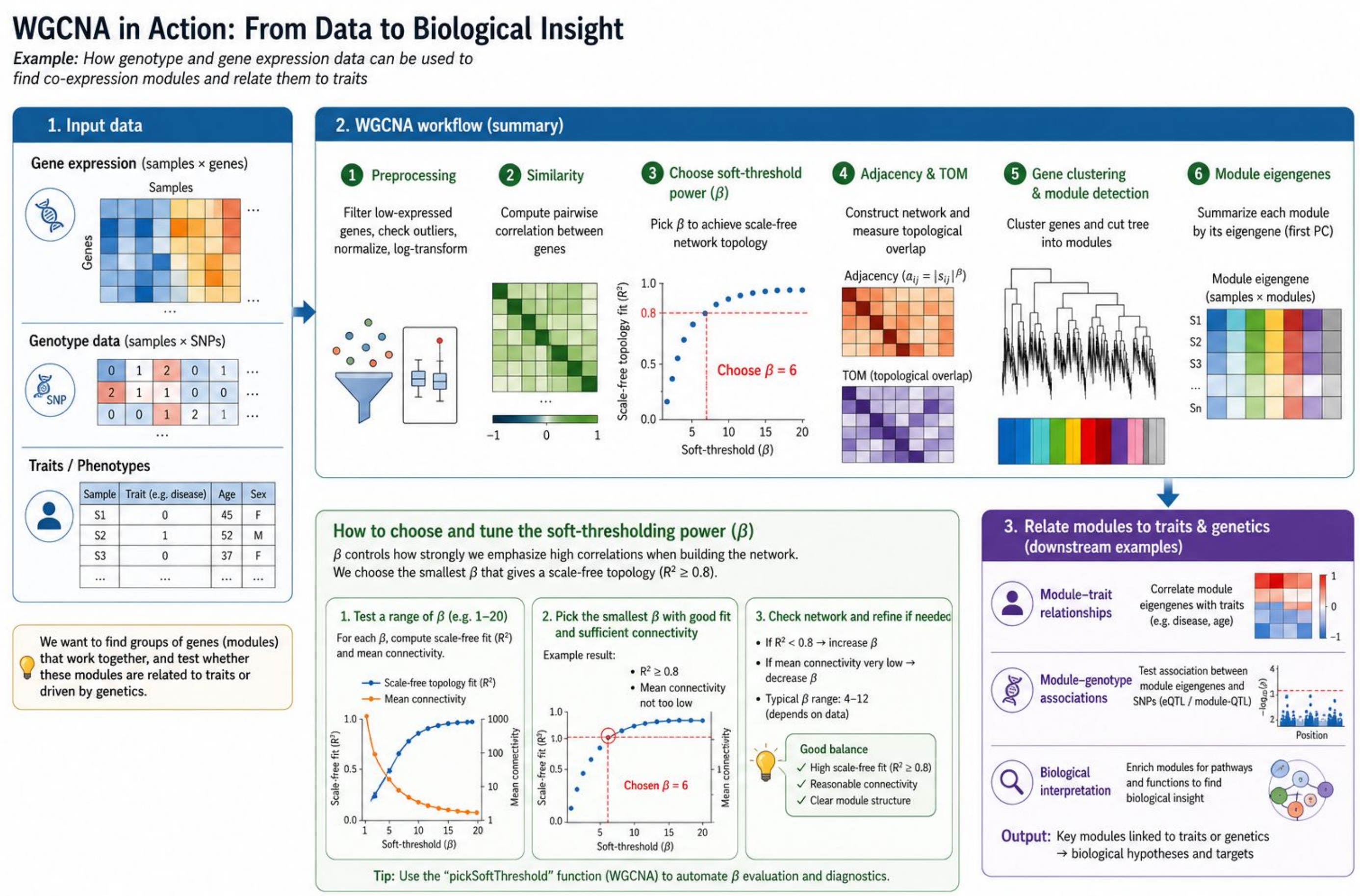
Biological insight
The Brown module is strongly linked to disease and to SNP1 genotype.

SNP1
↓
Brown module (Gene7, Gene8)
↓
Disease

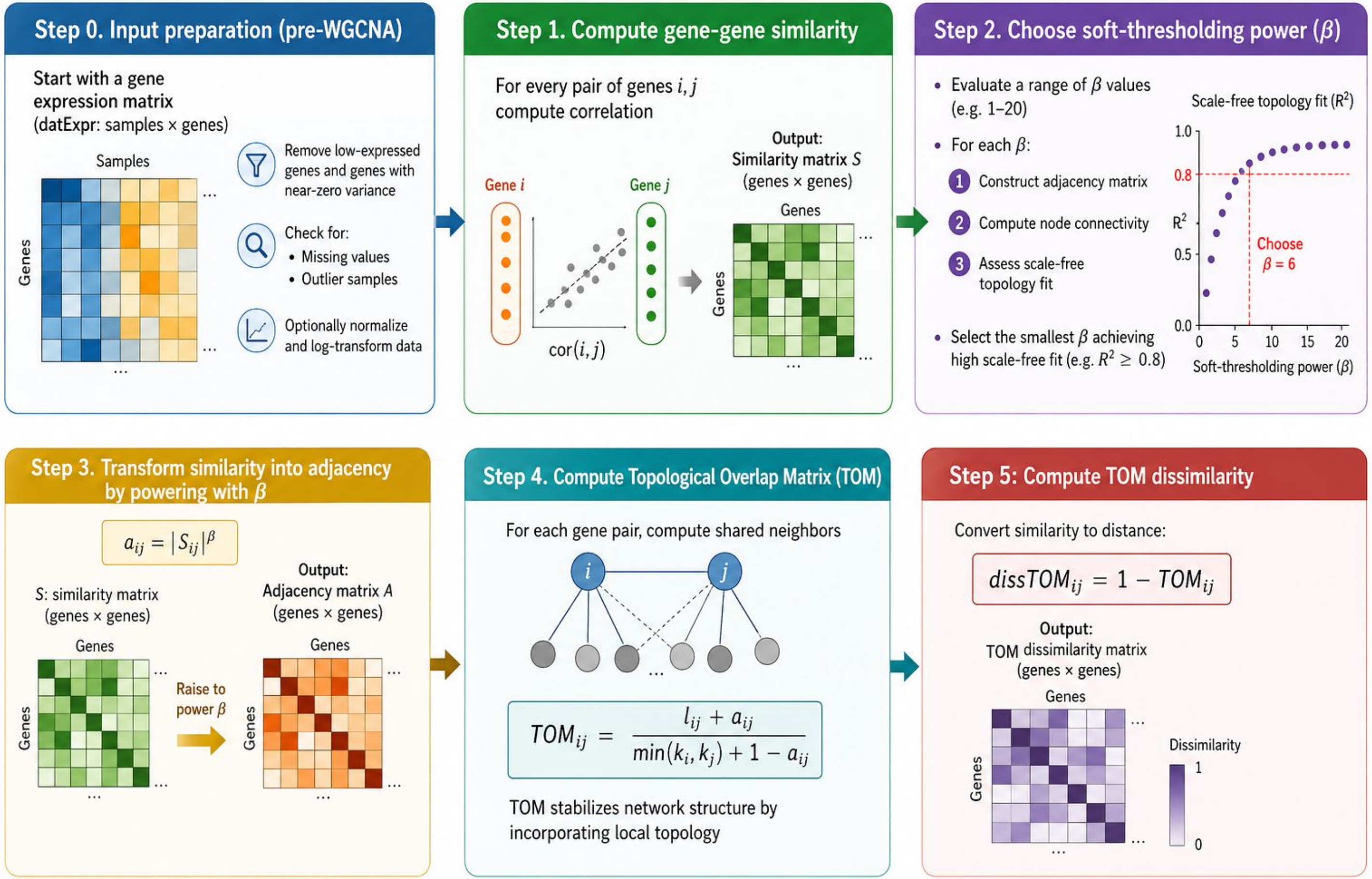
Why it's useful
- ✓ Reduces noise by using modules instead of individual genes.
- ✓ Finds biologically meaningful groups.
- ✓ Links gene networks to traits and genetics.
- ✓ Helps generate targets and hypotheses.

Take-home message: From a small dataset, WGCNA identified modules of co-expressed genes.
One module (Brown) was strongly linked to both disease and a genetic variant – pointing to a potential biological mechanism.

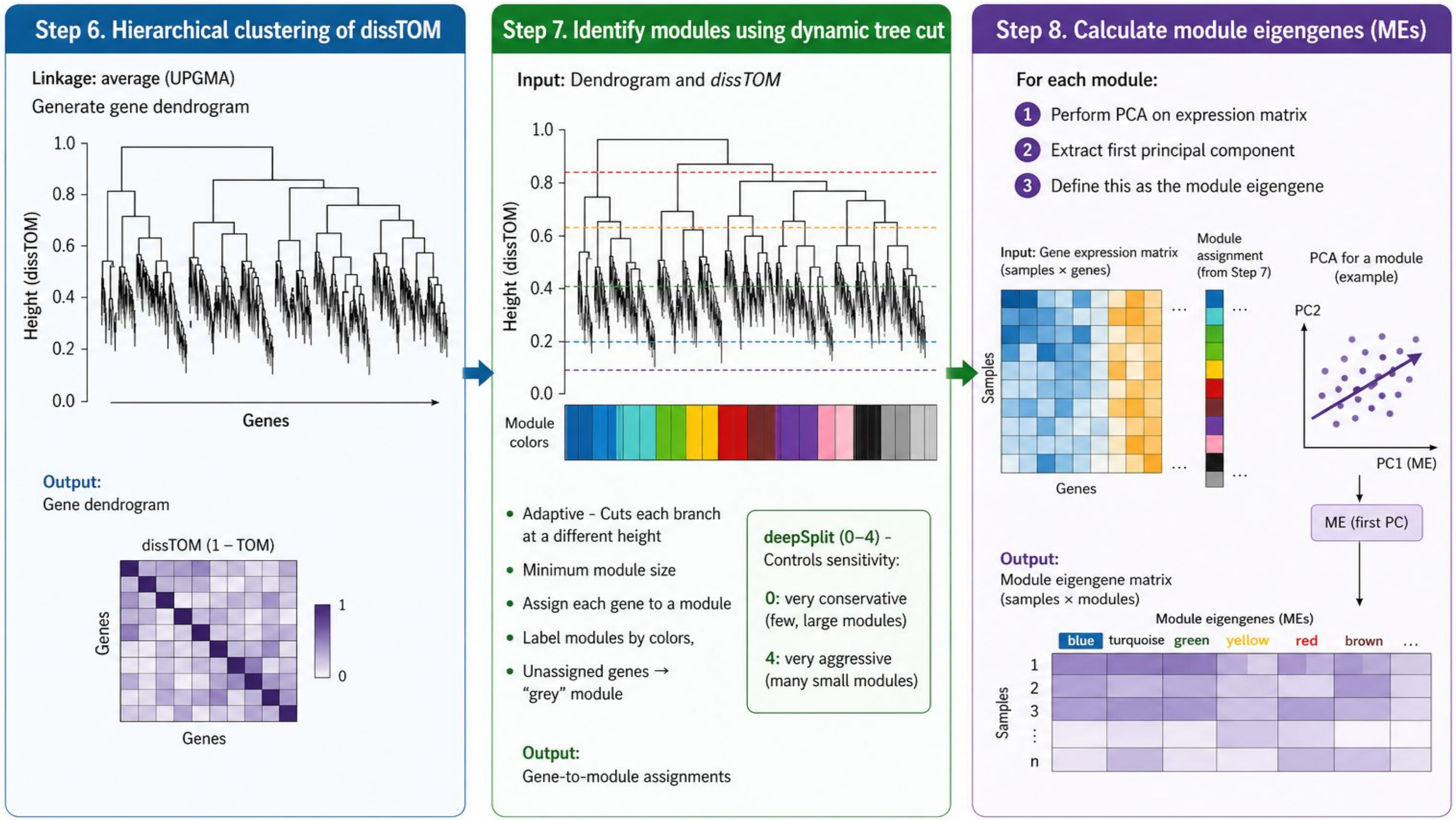
WGCNA - flow example



WGCNA - under the hood



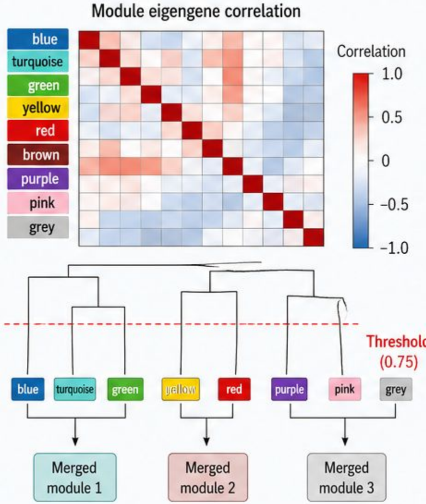
WGCNA - under the hood (2)



WGCNA - under the hood (3)

Step 9. Merge similar modules (optional but standard)

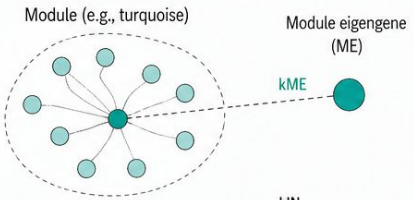
1. Compute correlations between module eigengenes
2. Cluster eigengenes
3. Merge modules with correlation above threshold (e.g. 0.75)



Output:
Merged module assignments and updated eigengenes

Step 10. Compute gene-module relationships

1. Calculate module membership (kME): correlation between gene expression and module eigengene
2. Compute intramodular connectivity (kIN): sum of connections of a gene to other genes in the same module
3. Identify hub genes (high kME and high kIN)



Gene	kME (module membership)	kIN (intramodular connectivity)	Hub gene?
Gene 1			✓
Gene 2			✓
Gene 3			✓
…			…
Gene n			

Output:
kME values and hub gene candidates per module

Step 11. Relate modules to external traits (downstream)

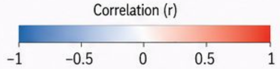
Correlate module eigengenes with:

- Phenotypes
- Clinical traits
- Genotypes (SNPs)

Identify biologically relevant modules

Module–trait relationships

	Trait 1 (e.g., age)	Trait 2 (e.g., status)	Trait 3 (e.g., BMI)	…
blue	−0.21	0.35	−0.12	…
turquoise	0.68	−0.41	0.22	…
green	−0.05	0.12	−0.30	…
yellow	0.10	−0.02	0.05	…
red	−0.32	0.61	−0.18	…
brown	0.02	−0.15	0.11	…
purple	−0.11	0.04	−0.07	…
grey	0.00	0.00	0.00	…
…				



Output:
Module–trait association statistics (correlations and p-values / FDR)

WGCNA - under the hood

- Step 0. Input preparation (pre-WGCNA)
  - Start with a gene expression matrix (datExpr: samples × genes)
  - Remove low-expressed genes and genes with near-zero variance
  - Check for: Missing values, Outlier samples
  - Optionally normalize and log-transform data
- Step 1. Compute gene-gene similarity
  - For every pair of genes i,j compute correlation
  - Output: Similarity matrix S (genes × genes)
- Step 2. Choose soft-thresholding power (β)
  - Evaluate a range of β values (e.g. 1–20)
  - For each β:
    - Construct adjacency matrix
    - Compute node connectivity
    - Assess scale-free topology fit
    - Select the smallest β achieving High scale-free fit (e.g. $R2$ ≥ 0.8)
- Step 3. Transform similarity into adjacency by powering with β
  - Output: Adjacency matrix (genes × genes)
- Step 4. Compute Topological Overlap Matrix (TOM)
  - For each gene pair, compute shared neighbors
  - TOM stabilizes network structure by incorporating local topology
- Step 5: Compute TOM dissimilarity
  - Convert similarity to distance: dissTOMij=1−TOMij

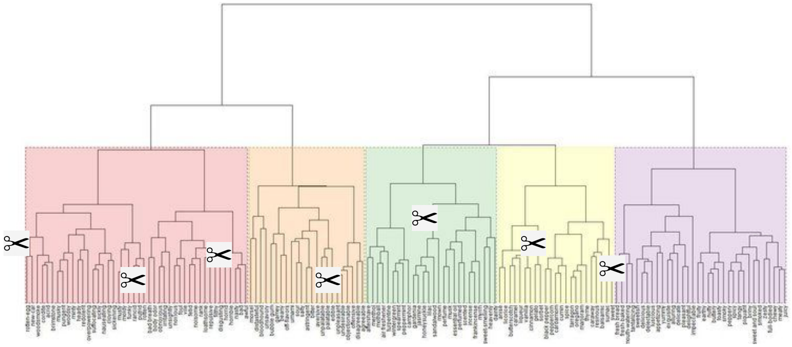
	Sample 1	Sample 2	Sample 3	Sample 4
Gene1	5.1	6.2	5.9	6.5
Gene2	4.9	6	6.1	6.3
Gene3	0.2	0.1	3.3	2.6
…	…	…	…	…
GeneK	0.3	0.7	3.3	2.6

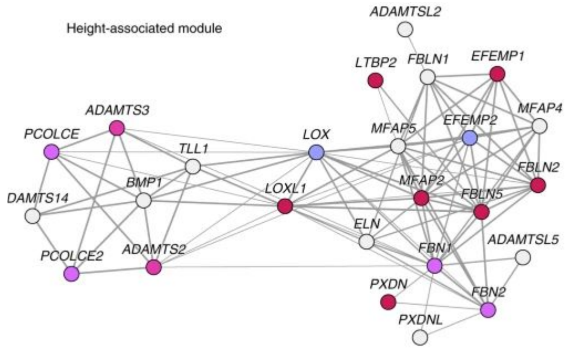
^β

	Gene 1	Gene 2	Gene 3	…	Gene K
Gene1	5.1	6.2	5.9	…	6.5
Gene2	4.9	6	6.1	…	6.3
Gene3	0.2	0.1	3.3	…	2.6
…	…	…	…	…	…
GeneK	0.3	0.7	3.3	…	2.6

WGCNA - under the hood (2)

- Step 6. Hierarchical clustering of dissTOM
  - Linkage: average (UPGMA)
  - Generate gene dendrogram
  - Output: Gene dendrogram
- Step 7. Identify modules using dynamic tree cut
  - Adaptive - Cuts each branch at a different height
  - Minimum module size
  - Assign each gene to a module
  - Label modules by colors, Unassigned genes → “grey” module
  - deepSplit (0–4) - Controls sensitivity:
    - 0: very conservative (few, large modules)
    - 4: very aggressive (many small modules)
  - Input: Dendrogram and dissTOM
  - Output: Gene-to-module assignments
- Step 8. Calculate module eigengenes (MEs)
  - For each module:
  - Perform PCA on expression matrix
  - Extract first principal component
  - Define this as the module eigengene
  - Output: Module eigengene matrix (samples × modules)



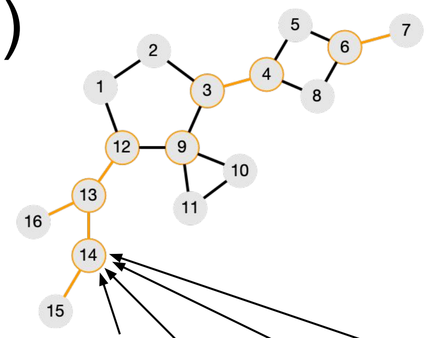


	ME0	ME1	ME2	ME3
sample				
sample_1	0.063365	-0.044246	-0.013674	0.057695
sample_2	0.114630	0.120750	-0.035715	0.032574
sample_3	0.012099	0.025344	0.069821	-0.042790
sample_4	-0.039167	-0.042957	-0.083667	0.007453
sample_5	-0.039167	-0.027502	-0.082489	-0.042790
...	...	...	...	...
sample_296	-0.028914	-0.049100	0.000210	-0.042790

* Choobdar et al. Assessment of network module identification across complex diseases

WGCNA - under the hood (3)

- Step 9. Merge similar modules (optional but standard)
  - Compute correlations between module eigengenes
  - Cluster eigengenes
  - Merge modules with correlation above threshold (e.g. 0.75)
  - Output: Merged module assignments and updated eigengenes
- Step 10. Compute gene-module relationships
  - Calculate module membership (kME):
  - Compute intramodular connectivity
  - Identify hub genes
  - Output: kME values and hub gene candidates
- Step 11. Relate modules to external traits (downstream)
  - Correlate module eigengenes with:
    - Phenotypes
    - Clinical traits
    - Genotypes (SNPs)
  - Identify biologically relevant modules
  - Output: Module-trait association statistics



sample	ME0	ME1	ME2	ME3
sample_1	0.063365	-0.044246	-0.013674	0.057695
sample_2	0.114630	0.120750	-0.035715	0.032574
sample_3	0.012099	0.025344	0.069821	-0.042790
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sample_5	-0.039167	-0.027502	-0.082489	-0.042790
...	...	...	...	...
sample_296	-0.028914	-0.049100	0.000210	-0.042790

sample	ASD	Age	Sex	IQ
sample_1	1	8.7	0	96
sample_2	0	11.4	1	134

WGCNA summary

Expression matrix
↓
Gene-gene correlation
↓
Soft thresholding (β)
↓
Adjacency matrix
↓
Topological Overlap Matrix
↓
Hierarchical clustering
↓
Dynamic tree cut
↓
Modules
↓
Module eigengenes
↓
Trait association

WGCNA - correlations

- Raw correlations are too dense and noisy:
- Cor = 0.2 vs 0.8 → both are “connected” if you threshold
- Hard thresholds lose information
- WGCNA instead reweights correlations non-linearly, making:
  - Strong correlations much stronger
  - Weak correlations close to zero
- That non-linearity is controlled by power (β): a_ij = |cor(x_i, x_j)|^β
  - soft-thresholding exponent (β) - tunable parameter
  - controls how strongly gene–gene correlations are converted into network connection strengths (adjacencies)

Biological Rationale

- Biological networks (gene regulation, protein interactions) tend to be scale-free:
  - Many genes have few connections
  - Few hub genes have many connections
- Formally, node degree distribution follows:

$$P(k) \sim k^{(-\gamma)}$$

- The power β is chosen so that the resulting network approximately satisfies this property.

Selecting Beta

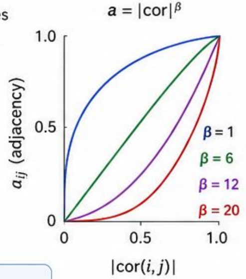
β controls how concentrated connectivity becomes — and hubs emerge from that concentration.

1 Mechanism: $a_{ij} = |\mathrm{cor}(i,j)|^{\beta}$ (non-linear filter)

The adjacency $a_{ij} = |\mathrm{cor}(i,j)|^{\beta}$ raises absolute correlations to the power β.

cor = 0.9 survives well
$0.9^6 \approx 0.53$

cor = 0.4 is crushed
$0.4^6 \approx 0.004$



As β grows, weak edges → 0 while strong edges stay near 1.

Each gene's connectivity $k_i = \sum_j a_{ij}$ collapses except where it has a few genuinely strong correlations.

The degree distribution becomes heavy-tailed — most genes have low k, a small set keeps high k. That's the definition of a hub structure.

2 Soft thresholds

β	mean k	max k	max/mean (hub-ness)
1	49.0	64.0	1.3
6	4.87	13.3	2.7
12	0.68	3.1	4.6
20	0.078	0.56	7.1

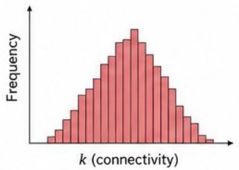
max/mean is a crude "hub-ness" index: higher β makes a few genes carry disproportionately more connections than the average — that's hub emergence.

Scale-free fit R^2 peaks where this concentration first follows a power law.

3 Trade-off

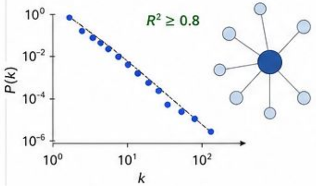
β too low (dense)

Network is dense, k is roughly uniform → no distinguishable hubs.



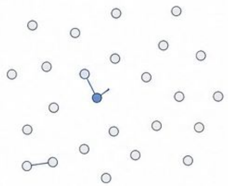
β optimal ($R^2 \geq 0.8$)

Clean scale-free topology, a small set of well-defined hubs per module.



β too high (sparse/fragmented)

Network fragments, even moderate correlations vanish → modules dissolve, " hubs" become singletons.



4 For hub gene identification specifically

WGCNA defines hubs by intramodular connectivity k_{IN} (or k_{ME}, correlation with module eigengene).
k_{IN} is computed after β is applied, so:

Higher β ⇒ k_{IN} is dominated by the genes' strongest within-module correlations → the gene ranking sharpens, fewer ties, hubs are clearer.

Lower β ⇒ k_{IN} values cluster together → many genes look "kind of hubby," none stand out.

So β doesn't change which gene is a hub by correlation, but it changes how cleanly hubs separate from non-hubs in the network metric.

Choose β where scale-free fit ($R^2 \geq 0.8$) is high and the network is still connected.
That gives you a clean, interpretable hub structure and sharp hub gene rankings.

Interpretation of β

- Think of β as a contrast knob:
  - Low β → blurry image (everything connected)
  - High β → high-contrast image (only strong edges survive)
- Too low:
  - Modules merge
  - Noise dominates
- Too high:
  - Network fragments
  - Loss of biologically meaningful structure

- Summary: power (β) is the soft-thresholding exponent that non-linearly transforms gene-gene correlations into network connection strengths so that the resulting network approximates scale-free topology and emphasizes biologically meaningful, strongly co-expressed gene relationships

Signed vs Unsigned Networks

- Unsigned:
  - $a_{ij} = |cor_{ij}|^\beta$
  - Positive and negative correlations treated equally
  - Useful when direction is irrelevant
- Signed (often recommended):
  - $a_{ij} = ((1 + cor_{ij})/2)^\beta$
  - Negative correlations → near zero adjacency
  - Modules represent positively co-expressed genes
  - Easier biological interpretation
- Signed networks often require lower β values

WGCNA inputs and outputs

Category	Object	Dimension
Input	Expression matrix	samples × genes
Optional input	Traits / genotypes	samples × traits
Core output	Module labels	genes
Core output	Module eigengenes	samples × modules
Core output	kME	genes × modules
Intermediate	TOM	genes × genes

Hub-gene identification — Module Membership (kME) and Gene Significance (GS)

A hub gene is one that is both:

1 Central in its module

High correlation with the module eigengene (Module Membership, also called kME or signed eigengene-based connectivity)

2 Biologically relevant

High correlation with a trait of interest (Gene Significance, GS)

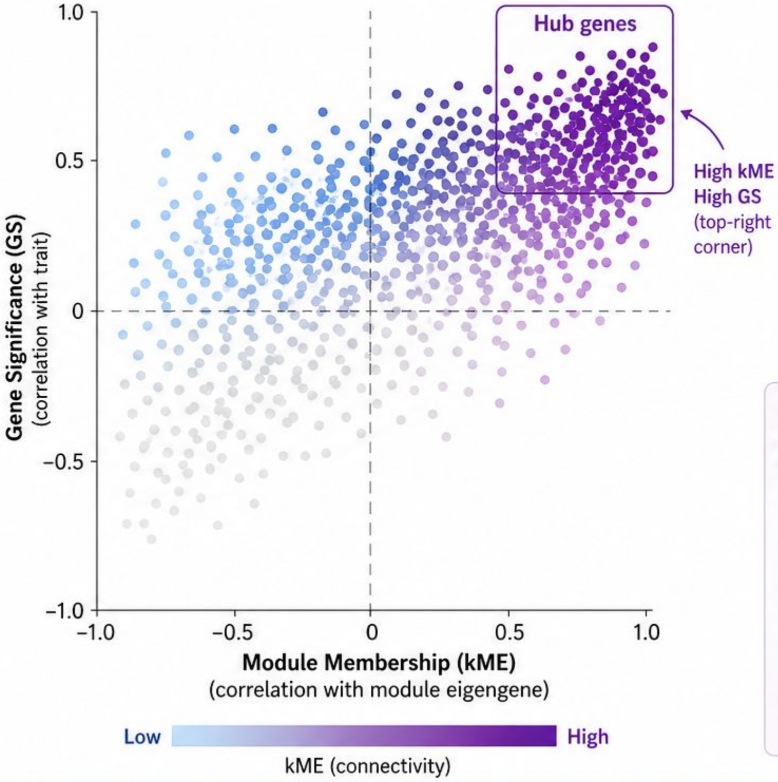
Plotting GS vs kME inside each module surfaces the genes in the upper-right corner: highly connected within the module and aligned with the phenotype.

Prime candidates for follow-up

- Functional validation
- Drug targets
- Biomarker panels

Example: Blue module

Each point = one gene



kME (Module Membership)

Correlation between a gene's expression profile and the module eigengene.
Measures how central the gene is in the module.

GS (Gene Signifcance)

Correlation between a gene's expression profile and the trait/phenotype.
Measures how relevant the gene is to the trait.

Example hub genes (upper-right)

Gene	kME	GS
SYT1	0.89	0.78
SYN1	0.86	0.72
GRIN2A	0.85	0.69
CAMK2A	0.83	0.71
DLG4	0.81	0.66
...	...	...

Highly connected & trait-aligned → prime candidates

Take-home message

High kME + High GS = hub genes that are both central to the module and relevant to the phenotype.
Focus on the upper-right corner in each module's GS vs kME plot.

Validate Target Biomarker

WGCNA R implementation

The 4 critical knobs at network construction (blockwiseModules / pickSoftThreshold)

Parameter	Effect	Tune via	Typical
power (β)	Soft-thresholding exponent: a_ij = |cor|^β	pickSoftThreshold() — smallest β with $R^2 \geq 0.8$	6 unsigned / 12 signed
networkType	unsigned | signed | signed hybrid	Use SIGNED when direction matters (most biology)	signed
TOMType	Topological overlap flavor; should match networkType	"signed" with signed network	signed
corType	pearson | bicor	bicor (biweight midcorrelation) is robust to outliers	bicor

pickSoftThreshold workflow

• Scan β = 1..20; plot R^2 vs β AND mean k vs β
• Pick smallest β with $R^2 \geq 0.8$ AND mean connectivity not collapsed near 0
• If no β reaches 0.8 → fall back to default (6 / 12) and document it

Signed vs unsigned in one sentence

• Unsigned merges co-up and co-down genes into one module → coarser modules
• Signed keeps positive co-expression only → biologically cleaner modules

WGCNA R implementation - module parameters

Module detection (blockwiseModules / cutreeDynamic)

Parameter	Range	Effect	Typical
minModuleSize	≥ 5	Smaller → many fine modules; larger → few coarse	30 (large) / 10–20 (small)
deepSplit	0–4	0 = conservative, 4 = aggressive splits	2
mergeCutHeight	0.0–1.0	Cut on ME-dendrogram; 0.25 ⇔ cor(ME,ME) > 0.75	0.25
pamRespectsDendro	TRUE/FALSE	Reassign unassigned genes to closest module	FALSE (cleaner)
maxBlockSize	≥ 5000	Memory cap; chunked clustering for large transcriptomes	5000–20000

Tuning workflow (in order)

1. Filter to top-variable expressed genes (≈ 5–15K)
2. pickSoftThreshold → choose β
3. Run blockwiseModules with defaults
4. Inspect dendrogram + module sizes
5. Iterate: tune minModuleSize, deepSplit, mergeCutHeight
6. Validate via module preservation Z-summary on a held-out cohort

Rule of thumb: tune β (controls hub structure) and mergeCutHeight (controls granularity) first.

ENRICHMENT: MAKING SENSE OF WGCNA MODULES

Enrichment is what moves WGCNA output from "we found gene clusters" to "we found a synaptic-signalling program disrupted in disease"

WHY?

1. A module is just a co-expressed gene list – needs a label
2. Enrichment names it: "synaptic", "cell cycle", "immune"
3. Connects the network to pathways, drug targets, literature

WGCNA OUTPUT

AFTER ENRICHMENT

Synaptic signalling program

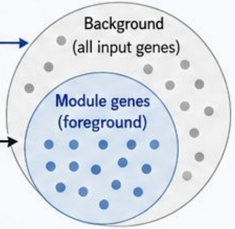
(disrupted in disease)

TWO METHODS

1. ORA (OVER-REPRESENTATION, FISHER EXACT)

Foreground = module genes

Background = WGCNA input genes



	In gene set	Not in gene set
In module (foreground)	a	b
Not in module (but in background)	c	d

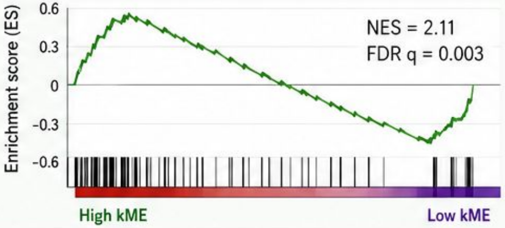
Fisher exact test → p-value

Fast, simple, ignores ranking
Good for concise lists and strong, discrete signals

2. RANKED GSEA

	Gene	kME
1	SYT1	0.92
2	SNAP25	0.88
3	NRXN1	0.79
4	RIMS1	0.76
	...	...
n-2	PPP1R1B	-0.62
n-1	MALAT1	-0.70
n	NEAT1	-0.85

Rank module genes by kME (network centrality)



Captures gradient signal, not just membership
More sensitive to subtle, coordinated effects

GENE-SET LIBRARIES

GO Biological Process (GO:BP)
Broad mechanisms and biological functions (e.g., synaptic signalling, axon guidance, apoptosis)

KEGG / Reactome Curated pathways
Well-annotated signaling and metabolic pathways (e.g., MAPK pathway, glutamatergic synapse)

MSigDB Hallmark (50 signatures)
Condensed, non-redundant biological states (e.g., HALLMARK SYNAPTIC SIGNALING, E2F TARGETS)

Cell-type panels (PanglaoDB, BRETIGEA)
Cell-type and cell-state gene sets – MANDATORY for bulk RNA-seq to capture cellular composition shifts

ENRICHMENT TURNS MODULES INTO MEANING: It labels, contextualizes, and connects your network to biology, disease, and therapeutics.